

SATOSHIFLOW

Bias-free BTC/USD strategy backtest

PROVISIONAL DATASET: The official 2019-2023 challenge file is absent. This report evaluates the repository's 2018-2022 fallback data and does not claim official results.

1. Objective and dataset

The objective is a reproducible BTC/USD trend strategy with strict next-candle execution, 0.15% brokerage on every entry and exit transaction, and daily mark-to-market accounting. The selected file is **btc_18_22_1d.csv** (1,826 rows), spanning 2018-01-01 through 2022-12-31. Its SHA-256 is

32814cd9f3b83dc79a0092c1f1c24bd988dad4c4b230dcd421401025ac660748.

2. Final out-of-sample metrics

Metric	Verified value
Evaluation period	2022 provisional out-of-sample
Initial / final equity	\$1,000.00 / \$1,210.46
Net return	21.05%
Sharpe ratio	0.709
Maximum drawdown	22.51%
Win rate	50.00%
Completed trades	6
Total brokerage	\$20.42
Buy-and-hold return	-64.21%

Gross profit before brokerage was **\$230.88**; net profit after **\$20.42** brokerage was **\$210.46**. There were 6 entries, 6 exits, and 0 reversals.

3. Strategy and formulas

The fixed strategy enters a 30-day Donchian breakout only when ADX is at least 20, directional movement agrees with the breakout, and close is on the correct side of the 200-day EMA. A $2.5 \times \text{ATR}$ trailing stop exits the position; an opposite confirmed breakout may reverse it. Parameters were selected from a 16-combination grid using only 2019-2021 yearly results, ranked by median then worst-year Sharpe. The 2022 fallback test was excluded from selection.

True Range: $\max(H-L, |H-C_{\text{prev}}|, |L-C_{\text{prev}}|)$. **ATR:** Wilder recursive average of True Range. **+DMI/-DM:** the larger positive directional high/low move; **DI:** $100 \times \text{Wilder}(\text{DM}) / \text{ATR}$. **DX:** $100 \times |DI+ - DI-| / (DI+ + DI-)$; **ADX:** Wilder average of DX.

EMA: $\alpha \times \text{close} + (1-\alpha) \times \text{prior EMA}$, where $\alpha = 2/(\text{period}+1)$. Donchian levels use shifted prior highs and lows, so the breakout candle is excluded.

4. Timing, fees, and accounting

A decision is formed only after candle t is complete and is queued for execution at candle $t+1$ open. Entry notional fully deploys available equity after reserving the entry fee. The engine charges 0.15% on entry notional and 0.15% on exit notional. Reversals close the old trade and open the new one, charging both transactions. Any final open position is force-closed at the last close.

Daily equity equals cash plus signed BTC quantity times daily close. Daily returns are $\text{equity}(t)/\text{equity}(t-1)-1$. Sharpe is $\sqrt{365}$ times mean daily return divided by sample standard deviation. Drawdown is $\text{equity}/\text{running maximum}-1$; the report states its largest magnitude as a positive percentage. Win rate uses completed trades only.

5. Mark-to-market equity

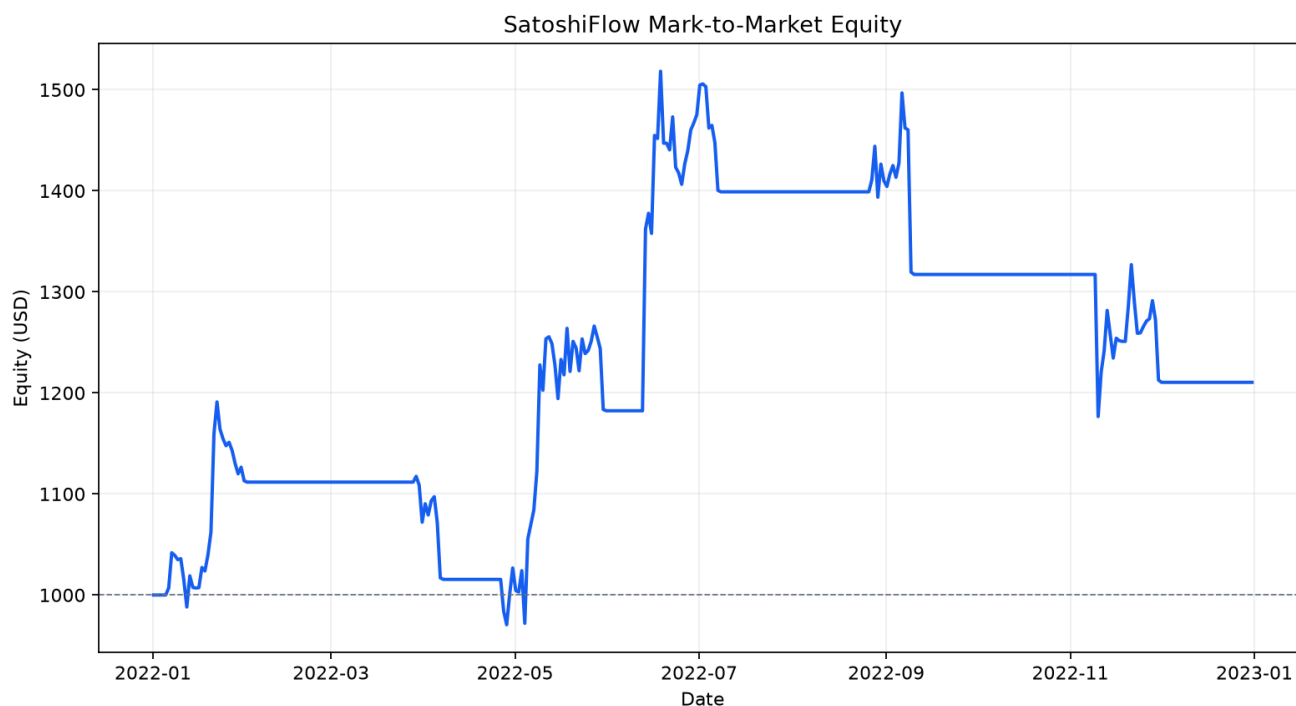


Figure 1. Daily equity after unrealized P&L; and transaction fees.

6. Drawdown and robustness

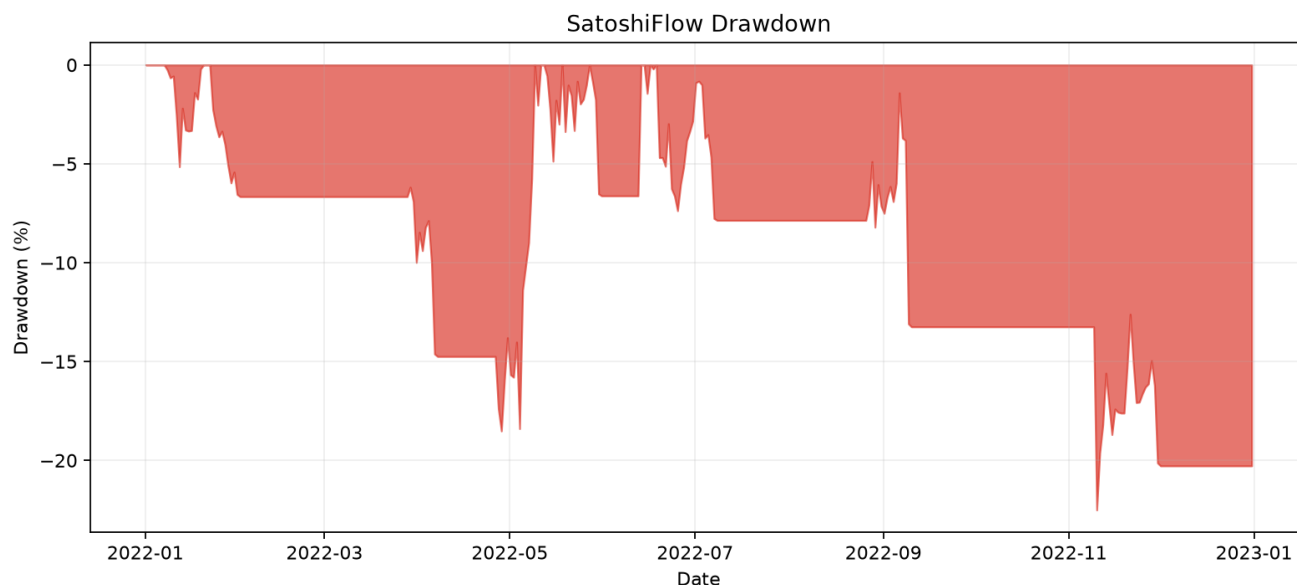


Figure 2. Peak-to-trough drawdown from daily mark-to-market equity.

7. Integrity checks

LOOKAHEAD CHECK: PASS. Signals and every used indicator are invariant when future suffixes are appended. Checks compare all bars through several cutoffs, not only trade bars.

NEXT-BAR EXECUTION: PASS. A fixture proves a target formed on t enters at $t+1$ open. **REPRODUCIBILITY CHECK: PASS.** Two complete runs produce identical signals, trades, equity, final equity, Sharpe, drawdown, and win rate.

8. Limitations

The official BTC_2019_2023_1d.csv training file is missing, so 2023 cannot be evaluated and all results are provisional. The fallback file may differ in exchange, timezone, price source, or candle construction. The simulator includes brokerage but not bid-ask spread, slippage, funding, borrow constraints, latency, or market impact. Six trades are too few to infer stable live performance. Re-run the documented command with the official file before final external submission.

Source of truth: results/metrics.json generated by main.py. Report values are loaded programmatically from that file.